The Derivative from First Principles

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The definition of the derivative**

Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

$\frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} = \frac{2xh + h^2 + 3h}{h} = 2x + h + 3 \to 2x + 3.$

2 **[Foundational]** • **Target: The definition of the derivative**

Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).

$\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x+h} + \sqrt{x}} \to \frac{1}{2\sqrt{x}}.$

3 **[Foundational]** • **Target: The definition of the derivative**

The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.

a Find the slope of the secant line. $\frac{9-1}{3-1} = 4.$

b Find $f'(2)$ using the derivative, and compare it to the secant slope. $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

(a) $\frac{9-1}{3-1} = 4.$; (b) $f'(x) = 2x$, so $f'(2) = 4$ — for this quadratic, the secant slope between $x = 1$ and $x = 3$ happens to equal the instantaneous slope at the midpoint $x = 2$.

4 **[Foundational]** • **Target: Differentiability**

Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.

The left-hand derivative is $\lim_{h \to 0^-} \frac{|h|}{h} = -1$ and the right-hand derivative is $\lim_{h \to 0^+} \frac{|h|}{h} = 1$. Since these disagree, $f'(0)$ does not exist.

5 **[Foundational]** • **Target: Differentiability**

State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

Differentiability implies continuity (a corner or asymptote would break the limit needed for the derivative). The converse is false — $f(x) = |x|$ is continuous at 0 but not differentiable there.

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6 [Foundational] • Target: More derivatives from the definition

Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).

$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x-(x+h)}{hx(x+h)} = \frac{-h}{hx(x+h)} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$.

7 [Foundational] • Target: More derivatives from the definition

Use the limit definition to find $f'(x)$ for $f(x) = x^3$.

$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, so $\frac{f(x+h)-f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \to 3x^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Interpreting the derivative

A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.

$h'(2)$ represents the instantaneous vertical velocity of the ball at $t = 2$ seconds. $h'(2) = -10(2) + 20 = 0$ m/s — the ball is momentarily at rest, at its peak height.

9 [Intermediate] • Target: Point-derivatives from the definition

Use the limit definition $f'(a) = \lim_{h \to 0} \frac{f(a+h)-f(a)}{h}$ to find $f'(4)$ for $f(x) = x^2 - 3x$.

$f(4) = 4$. $f(4+h) = (4+h)^2 - 3(4+h) = 16 + 8h + h^2 - 12 - 3h = 4 + 5h + h^2$.
$\frac{f(4+h)-f(4)}{h} = \frac{5h + h^2}{h} = 5 + h \to 5$: $f'(4) = 5$.

10 [Intermediate] • Target: Point-derivatives from the definition

Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x)-f(a)}{x-a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

$f(2) = 7$. $\frac{x^3 - 1 - 7}{x-2} = \frac{x^3 - 8}{x-2} = \frac{(x-2)(x^2+2x+4)}{x-2} = x^2 + 2x + 4 \to 4 + 4 + 4 = 12$: $f'(2) = 12$.

11 [Intermediate] • Target: The derivative as the slope of the tangent line

For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.

$f(3) = 9 - 12 + 1 = -2$. $f'(3) = 2(3) - 4 = 2$. Tangent line: $y - (-2) = 2(x - 3)$, i.e. $y = 2x - 8$.

12 [Intermediate] • Target: The derivative as the slope of the tangent line

Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.

$f'(0) = 2(0) = 0$. Since the slope of the tangent line at $x = 0$ is 0, the tangent line is horizontal there — consistent with $x = 0$ being the vertex (minimum point) of the parabola.

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13 [Intermediate] • Target: The derivative function versus the derivative at a point

Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.

$f'(x) = 2x$ is a FUNCTION giving the slope of the tangent line at any point x. $f'(3) = 2(3) = 6$ is a single NUMBER — the specific slope at the point $x = 3$. The function $f'(x)$ can be evaluated at any x to get such a number.

14 [Intermediate] • Target: Estimating a derivative from a table

A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

$f'(2) \approx \frac{f(2.1) - f(1.9)}{2.1 - 1.9} = \frac{9.41 - 8.61}{0.2} = \frac{0.80}{0.2} = 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 11.01 Sequences and Series

Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = \frac{0}{h} = 0$ for all $h \neq 0$, so $f'(x) = \lim_{h \to 0} 0 = 0$.

16 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 12.01 Convergence Tests

Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.

$\frac{f(x+h) - f(x)}{h} = \frac{m(x+h) + b - (mx + b)}{h} = \frac{mh}{h} = m$ for all $h \neq 0$, so $f'(x) = \lim_{h \to 0} m = m$.

17 [Challenging] • Target: Interpreting the derivative • Cross-Lesson: 12.02 Power Series and Taylor Series

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$?

The integral of $f(x) = x^2$ is $\frac{x^3}{3}$. Evaluating this from 0 to 3 gives: $\left[\frac{x^3}{3}\right]_0^3 = \frac{3^3}{3} - \frac{0^3}{3} = \frac{27}{3} - 0 = 9$. Therefore, the value of the definite integral is 9.

18 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Find the directional derivative of $f(x, y) = \sqrt{x^2 + y^2 + 4}$ at $(1,\ 3)$ in the direction of $\langle 4,\ -3 \rangle$.

$ablaf(1,\ 3) = \langle \frac{\sqrt{14}}{14}, \frac{3\sqrt{14}}{14} \rangle$ and the unit vector is $\langle \frac{4}{5},\ -\frac{3}{5} \rangle$. $D_{\mathbf{u}} f = ablaf \cdot \mathbf{u} = -\frac{\sqrt{14}}{14}$.

19 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate $\int_0^{\pi} e^x \sin(2x)\, dx$.

Integrate by parts twice; the original integral reappears, so solve for it:
$\int e^x \sin(2x)\, dx = \frac{(\sin(2x) - 2\cos(2x))e^x}{5}$. From 0 to π: $\frac{2}{5} - \frac{2e^{\pi}}{5}$.

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20 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 15.01 Double Integrals

If $y = x^{\sin(x)}$, find $\frac{dy}{dx}$ at $x = 4$.

Take logs: $\ln(y) = \ln(x)\sin(x)$, differentiate, and multiply by y. Substituting $x = 4$ gives $\frac{8\ln(2)\cos(4) + \sin(4)}{2^{2-2\sin(4)}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Limit Definition of the Derivative

State the standard limit definition of $f'(x)$.

a. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$ ✓

b. $f'(x) = \lim_{h\to 0} \frac{f(x) - f(h)}{h}$

c. $f'(x) = \frac{f(x+h) - f(x)}{h}$

d. $f'(x) = \lim_{x\to 0} \frac{f(x+h) - f(x)}{x}$

The derivative represents the instantaneous rate of change defined by the limit of the difference quotient as $h \to 0$.

22 [Foundational] • Target: The Limit Definition of the Derivative

Use first principles to find the derivative of $f(x) = 5x^2 - 3x$.

a. $10x - 3$ ✓

b. $10x$

c. $5x - 3$

d. $10x + 3$

$\lim_{h\to 0} \frac{5(x+h)^2 - 3(x+h) - (5x^2 - 3x)}{h} = \lim_{h\to 0} \frac{10xh + 5h^2 - 3h}{h} = \lim_{h\to 0} (10x + 5h - 3) = 10x - 3.$

23 [Foundational] • Target: The Limit Definition of the Derivative

Use the alternative limit definition $f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

a. 12 ✓

b. 8

c. 6

d. 4

$\lim_{x\to 2} \frac{x^3 - 8}{x - 2} = \lim_{x\to 2} \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = 2^2 + 2(2) + 4 = 12.$

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24 [Foundational] • Target: First Principles for Radicals and Reciprocals

Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$.

a. $-1/x^2$ ✓

b. $1/x^2$

c. $-1/x$

d. $-2/x^2$

$\lim_{h\to 0} \frac{1/h \cdot (x-(x+h))}{x(x+h)} = \lim_{h\to 0} \frac{-h}{hx(x+h)} = \lim_{h\to 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}$.

25 [Foundational] • Target: First Principles for Radicals and Reciprocals

Find $f'(x)$ from first principles for $f(x) = \sqrt{x}$ $(x > 0)$.

a. $\frac{1}{2\sqrt{x}}$ ✓

b. $\frac{1}{\sqrt{x}}$

c. $-\frac{1}{2\sqrt{x}}$

d. $\frac{2}{\sqrt{x}}$

$\lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} = \lim_{h\to 0} \frac{(x+h)-x}{h(\sqrt{x+h}+\sqrt{x})} = \frac{1}{2\sqrt{x}}$.

26 [Foundational] • Target: First Principles for Radicals and Reciprocals

Evaluate $\lim_{h\to 0} \frac{(2+h)^5 - 32}{h}$ by recognizing it as a derivative.

a. 80 ✓

b. 40

c. 32

d. 16

This is $f'(2)$ for $f(x) = x^5$. Since $f'(x) = 5x^4$, $f'(2) = 5(2^4) = 5(16) = 80$.

27 [Foundational] • Target: Differentiability and Continuity

Which statement correctly describes the relationship between differentiability and continuity?

a. **Differentiability implies continuity, but continuity does not imply differentiability** ✓

b. Continuity implies differentiability

c. They are completely independent properties

d. A function is differentiable if and only if it is continuous

If f is differentiable at c, it must be continuous at c. However, functions like $f(x) = |x|$ are continuous at 0 but not differentiable there.

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28 [Foundational] • Target: Differentiability and Continuity

Why does $f(x) = |x - 3|$ fail to be differentiable at $x = 3$?

a. **The left-hand and right-hand derivative limits are unequal ($-1 \neq 1$) ✓**

b. $f(3)$ is undefined

c. f is discontinuous at $x = 3$

d. The tangent line is horizontal

Left derivative is $\lim_{h \to 0^-} \frac{-(3+h-3)}{h} = -1$. Right derivative is $+1$. Since they differ, a sharp corner exists and $f'(3)$ does not exist.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Differentiability and Continuity

At $x = 0$, the function $f(x) = x^{1/3}$ has which type of non-differentiability?

a. **Vertical tangent ($f'(x) \to \infty$) ✓**

b. Corner / sharp turn

c. Cusp

d. Removable discontinuity

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. As $x \to 0$, $f'(x) \to +\infty$. The tangent line is vertical, so f is not differentiable.

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30 [Intermediate] • Target: Geometric and Physical Interpretations

What geometric feature does the derivative $f'(a)$ represent?

a. **The slope of the tangent line to $y = f(x)$ at $x = a$** ✓

b. The area under the curve from 0 to a

c. The secant line slope between 0 and a

d. The curvature at $(a, f(a))$

The derivative $f'(a)$ is geometrically defined as the exact slope of the tangent line to the curve $y = f(x)$ at $(a, f(a))$.

31 [Intermediate] • Target: Geometric and Physical Interpretations

A particle moves along a line with position $s(t) = 4t^2 - 3t + 1$. What is its instantaneous velocity at $t = 2$?

a. 13 ✓

b. 11

c. 16

d. 10

$v(t) = s'(t) = 8t - 3$. At $t = 2$: $v(2) = 8(2) - 3 = 13$.

32 [Intermediate] • Target: Geometric and Physical Interpretations

Find the equation of the tangent line to $y = x^2 - 4x$ at the point where $x = 3$.

a. $y = 2x - 9$ ✓

b. $y = 2x - 3$

c. $y = -x$

d. $y = 4x - 15$

$y(3) = 9 - 12 = -3$. $y' = 2x - 4 \Rightarrow y'(3) = 2$. Line: $y - (-3) = 2(x - 3) \Rightarrow y = 2x - 9$.

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33 [Intermediate] • Target: Geometric and Physical Interpretations

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?

a. **4** ✓ **b.** 2

c. 3 **d.** 6

The slope of the tangent line to the graph of $f(x)$ at any point is given by its derivative, $f'(x)$. 1. Find the derivative of $f(x) = x^2 - 2x - 1$ using the power rule: $f'(x) = 2x - 2$ 2. Evaluate the derivative at the x -coordinate of point $P(3, 2)$, which is $x = 3$: $f'(3) = 2(3) - 2 = 6 - 2 = 4$ Thus, the slope of the tangent line at P is 4. Misconception Analysis: - 2 results from mistakenly reading the y-coordinate of point P as the slope, since $f(3) = 2$. - 3 results from taking the x-c.

34 [Intermediate] • Target: The definition of the derivative

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.

a. **1** ✓ **b.** 2

c. 3 **d.** 4

The equation of the tangent line to $f(x)$ at $x = 1$ is given as $y = 2x + c$. We are given that $f(1) = 3$. This means the point $(1, 3)$ lies on both the function $f(x)$ and its tangent line at $x = 1$. Since the point $(1, 3)$ lies on the tangent line $y = 2x + c$, we can substitute the coordinates into the equation: $3 = 2(1) + c$ $3 = 2 + c$ $c = 3 - 2$ $c = 1$ The derivative of the function at $x = 1$ is the slope of the tangent line, which is 2. This information is consistent with the problem but not direct.

35 [Intermediate] • Target: Differentiability

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

a. **2** ✓ **b.** 4

c. 8 **d.** 10

To find the partial derivative of $f(x, y) = x^2 + 2y^2$ with respect to x, we treat y as a constant and differentiate with respect to x. The derivative of x^2 with respect to x is $2x$, and the derivative of $2y^2$ (a constant with respect to x) is 0. Therefore, $\frac{\partial f}{\partial x} = 2x$. Now, we evaluate this partial derivative at the point $(1, 2)$. Substituting $x = 1$ into the expression for $\frac{\partial f}{\partial x}$: $\frac{\partial f}{\partial x}$ at $(1, 2) = 2 \times 1 = 2$. Th.

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36 [Intermediate] • Target: More derivatives from the definition

Tangent line slope to $y = \ln x$ at $x = e$:

a. 1

b. e

c. $1/e$ ✓

d. $\ln e$

Applying Applications of Derivatives principles, $f' = 1/x$; $f'(e) = 1/e$. Therefore, the verified correct option is $1/e$.

37 [Intermediate] • Target: Interpreting the derivative

The derivative of $f(x) = \frac{x^4 - 1}{x^2 + 1}$ at $x = 1$ using limit definition:

a. 1

b. **2** ✓

c. 3

d. 4

Simplify: $\frac{(x^2 - 1)(x^2 + 1)}{x^2 + 1} = x^2 - 1$. So $f'(x) = 2x$. At $x = 1$: $f'(1) = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

c. zero to zero

d. **positive to negative** ✓

f increasing then decreasing through c means c is a local maximum.

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39 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to x?

a. $2x - 5$ ✓

b. $x - 5$

c. $x^2 - 5$

d. $2x$

To find the derivative of the function $f(x) = x^2 - 5x + 6$, we apply the power rule for differentiation, which states that the derivative of ax^n is anx^{n-1}. The derivative of a constant is 0. Applying the power rule to each term: 1. The derivative of x^2 is $2x^{2-1} = 2x$. 2. The derivative of $-5x$ is $-5 \cdot 1 \cdot x^{1-1} = -5x^0 = -5$. 3. The derivative of the constant 6 is 0. Combining these, the derivative of $f(x)$ is $f'(x) = 2x - 5 + 0 = 2x - 5$.

40 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 15.01 Double Integrals

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?

a. **1** ✓

b. 0

c. 2

d. Does not exist

The limit of a function as x approaches a certain value is the value that $f(x)$ gets arbitrarily close to as x gets arbitrarily close to that value. Observing the graph, as x approaches 1 from both the left and the right, the value of $f(x)$ approaches 1. Therefore, $\lim_{x \to 1} f(x) = 1$.

41 [Challenging] • Target: Estimating a derivative from a table • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The slope of $y = 3x - 5$ at any point is:

a. -5

b. 0

c. **3** ✓

d. undefined

Applying Derivatives principles, Slope of a line $y = mx + b$ is $m = 3$. Therefore, the verified correct option is 3.

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42 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 2.01 Evaluating Limits

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. **Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$** ✓

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

First, we need to find the velocity, which is the first derivative of the position function $s(t) = f(t) = e^t \sin(t)$. Using the product rule, $v(t) = f'(t) = e^t \sin(t) + e^t \cos(t) = e^t(\sin(t) + \cos(t))$. The particle is momentarily at rest when $v(t) = 0$. Since $e^t > 0$ for all t, we need $\sin(t) + \cos(t) = 0$, which implies $\tan(t) = -1$. The general solutions for this are $t = \frac{3\pi}{4} + n\pi$, where n is an integer. For $t \geq 0$, the smallest positive value occurs when $n = 0$, so.

43 [Challenging] • Target: The Limit Definition of the Derivative • Cross-Lesson: 2.02 Continuity

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

a. $2x - 3y$

b. $2x - 3xy + 4y$

c. $-3x + 4y$ ✓

d. $2x - 3y + 4y$

To find the partial derivative of $f(x, y) = x^2 - 3xy + 2y^2$ with respect to y, we treat x as a constant. \\ \\ The derivative of x^2 with respect to y is 0, as x is considered constant. \\ \\ The derivative of $-3xy$ with respect to y is $-3x \cdot 1 = -3x$. \\ \\ The derivative of $2y^2$ with respect to y is $2 \cdot 2y = 4y$. \\ \\ Therefore, the partial derivative $\frac{\partial f}{\partial y}$ is $0 - 3x + 4y = -3x + 4y$.

44 [Challenging] • Target: First Principles for Radicals and Reciprocals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Find the slope of the tangent line to $r = \sin(2\theta)$ at $\theta = \frac{\pi}{3}$.

a. -1

b. $-\frac{5\sqrt{3}}{3}$

c. $\frac{\sqrt{3}}{5}$ ✓

d. $\sqrt{3}$

With $x = r\cos\theta$, $y = r\sin\theta$: $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$. At $\theta = \frac{\pi}{3}$ this is $\frac{\sqrt{3}}{5}$.

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45 [Challenging] • Target: Differentiability and Continuity • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Find the slope of the tangent line to $r = 2\sin(\theta) + 1$ at $\theta = \frac{2\pi}{3}$.

a. $-4 + 3\sqrt{3}$ ✓

b. -1

c. $-\frac{3\sqrt{3}}{11} - \frac{4}{11}$

d. $-\sqrt{3}$

With $x = r\cos\theta$, $y = r\sin\theta$: $\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$. At $\theta = \frac{2\pi}{3}$ this is $-4 + 3\sqrt{3}$.

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